

Uttkarsh Thakur

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SKILLS

Languages: Java, Python, TypeScript, JavaScript, SQL, C++

Backend: Spring Boot, Spring Web MVC, Spring Data JPA, Hibernate, REST API Design, Bean Validation, FastAPI, Flask, Node.js

Databases: PostgreSQL, MySQL, H2, MongoDB, Flyway Migrations, JPA/Hibernate ORM, Schema Design

Frontend: React.js, TypeScript, Tailwind CSS, Vite, HTML/CSS

Testing & Tools: JUnit 5, Mockito, AssertJ, Docker, Maven, Git & GitHub, Swagger/OpenAPI, Postman

AI/ML: RAG Pipelines, Semantic Search, FAISS, LangChain, Transformers, PyTorch, TensorFlow, Scikit-learn

PROJECTS

Billsync | Java 21, Spring Boot, Spring Data JPA, PostgreSQL, Flyway, React, TypeScript, Docker 2026

- Architected a full-stack expense-settlement platform exposing **16 REST endpoints** over a layered **controller-service-repository** design, with DTO boundaries, **Bean Validation**, and a single `@RestControllerAdvice` mapping every failure to a consistent **400/404/409** JSON contract.
- Engineered an exact-arithmetic split engine on **BigDecimal** using round-down plus deterministic remainder distribution, guaranteeing shares always sum to the expense total; validated by **brute-force sweeps** over every amount from 0.01 to 20.00 split 1–12 ways.
- Designed **SettlementPlanner**, a greedy **dual max-heap** debt-simplification algorithm running in **$O(n \log n)$** that bounds any group to **at most $n-1$ payments**, benchmarked against the direct debt graph and returning whichever plan is shorter.
- Modeled **6 normalized tables** with **Flyway**-versioned migrations and Hibernate in validate-only mode; eliminated **$N+1$ queries** via `@EntityGraph`, collapsing **22 SELECTs into 1** for a 20-member group.
- Achieved **73 automated tests** across 9 **JUnit 5/Mockito/AssertJ** suites covering service logic and **MockMvc** web slices, running on in-memory **H2** with zero Docker dependency.

CodeNav AI | FastAPI, React.js, LangChain, FAISS, Sentence-Transformers, OpenAI API, GitPython 2026

- Architected a full-stack **RAG** pipeline using **LangChain** and a **FAISS vector database**, enabling natural language Q&A over local codebases via **semantic search** with `all-MiniLM-L6-v2` embeddings.
- Engineered **AST-based parsing** to extract logical code units (functions/classes), preserving semantic boundaries to deliver more relevant retrieval context than naive fixed-size chunking.
- Optimized vector index loading via **persistent disk-based caching**, cutting load times from **>20s to <100ms** — a **99.5% reduction**; delivered through a **FastAPI** REST API and a **React.js** front-end.

Motion Sensing System | TensorFlow/Keras, VGG16, Flask, Streamlit, OpenCV, Python 2024

- Engineered a **Human Activity Recognition** system classifying real-time movements with a **VGG16** model fine-tuned via **transfer learning**, served through a **Flask** REST API with a **Streamlit** dashboard for live prediction.

RESEARCH & TALKS

Agreement Loops in Multi-Agent Systems | Seminar Talk, Literature Review 2026

- Surveyed foundational and recent papers on **consensus mechanisms** in **distributed multi-agent systems** and presented the findings as a **seminar talk**, synthesizing convergence criteria and open challenges in agent coordination.

EDUCATION

Bennett University

Aug. 2023 – May 2027

B.Tech in Computer Science and Engineering | **CGPA: 8.17/10**

Greater Noida, India

- Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Deep Learning, Neural Networks, System Design

DAV Public School

Graduated 2022

Senior Secondary Education (Class XII)

Ranchi, India

CERTIFICATIONS

Build Basic Generative Adversarial Networks (GANs) — DeepLearning.AI via Coursera

Apr. 2026

Neural Networks and Deep Learning — DeepLearning.AI via Coursera

Apr. 2025

Machine Learning: Clustering & Retrieval — University of Washington via Coursera

Nov. 2024

ACHIEVEMENTS

Ranked **101st** nationally at **Smart India Hackathon 2024**

CodeChef: 2-star rated | **LeetCode:** Contest rating **1500+**

Competed in **Hackaccino Hackathon**, Bennett University (2024)